

Prognosis of severe drug-induced acute interstitial nephritis requiring renal replacement therapy.

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Drug-induced acute interstitial nephritis (DAIN) is often associated with improved outcomes, whereas some patients may still progress to chronic kidney disease (CKD). The aim of this study was to evaluate the prognosis of patients with severe DAIN requiring renal replacement therapy (RRT) at baseline, and to explore the risk factors of progression to CKD. We performed a retrospective study of patients with severe DAIN confirmed by renal biopsies in our center over a 10 years period, all the patients received RRT at presentation. The clinical and pathological characteristics at baseline were recorded, and the outcomes (renal function recovered or progressed to CKD) during follow-ups were also evaluated. Univariate and multivariate logistic regression analysis were performed to identify the independent risk factors of progression to CKD. Seventy-two patients who met the inclusion criteria were enrolled, 13 patients (18.0%) progressed to CKD (GFR < 60 ml/min/1.73 m²) after at least 6 months of follow-up, the remaining 59 patients achieved a favorable renal function recovery. Compared with patients who achieved renal function recovery (recovery group), the patients progressed to CKD (progression group) were older and had longer interval from symptom onset to treatment with steroids. The peak serum cystatin C concentration was higher in progression group than recovery group. Higher score of interstitial fibrosis/tubular atrophy (IFTA) and more interstitial inflammatory cells infiltration were detected in renal tissue in progression group. According to multivariable analysis, higher peak cystatin C concentration (OR = 2.443, 95% CI 1.257, 4.746, $p = 0.008$), longer interval to treatment with corticosteroids (OR = 1.183, 95% CI 1.035, 1.352, $p = 0.014$) were independent risk factors of progression to CKD. The cutoff value of cystatin C concentration was 4.34 mg/L, at which the sensitivity and specificity were 76.9% and 89.3%, respectively; the cutoff value of interval to treatment with corticosteroids was 22.5 days, at which the sensitivity and specificity were 81.8% and 79.5%, respectively. Renal function was reversible in majority of patients with severe DAIN requiring RRT when early identification and treatment. Higher peak cystatin C concentration and longer interval to treatment with corticosteroids associated with worse renal prognosis.

Field Study and Correlative Studies of Factor IX Variant FIX-R338L in Participants Treated with Fidanacogene Elaparvovec.

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Fidanacogene elaparvovec, an adeno-associated virus-based gene therapy vector expressing the high-activity factor IX (FIX) variant FIX-R338L, is in development for hemophilia B. One-stage clotting (OS) assays and chromogenic substrate (CS) assays are commonly used to measure FIX-R338L variant activity. Data from ongoing trials suggest FIX activity varies between different OS and CS assays. To better understand FIX-R338L activity in clinical samples, an international multisite field study was conducted across a central laboratory and 18 local laboratories, using standard protocols, reagents, and instrumentation, with individual participant samples from a phase 1/2a study of fidanacogene elaparvovec. Unlike the wild-type FIX control, FIX-R338L activity was higher with the OS silica-based assay versus OS ellagic acid-based and CS assays. Variation in FIX activity was greater at the lowest activity levels. Activated FIX (FIXa) in plasma could result in higher OS assay activity or increased thrombin generation, which could overestimate FIX activity. However, FIXa was not detected in the participant samples, indicating that it was not contributing to the OS assay differences. Since individuals

on gene therapy may receive exogenous replacement FIX products, replacement products were spiked into patient plasma samples to target a therapeutic concentration. Exogenous FIX was additive to endogenous FIX-R338L, with no interference from FIX-R338L. These results demonstrate FIX-R338L activity can be measured with OS and CS assays in clinical laboratories and provide insight into assay variability when measuring FIX with endogenously produced FIX-R338L. The findings may help establish best practices for measuring FIX-R338L activity (Clinicaltrials.gov identifier: NCT02484092).

Fidanacogene Elaparvovec: First Approval.

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Fidanacogene elaparvovec (PrBEQVEZ™) is an adeno-associated viral (AAV) vector-based gene therapy developed by Spark Therapeutics (a subsidiary of Roche) and Pfizer (under a license from Spark Therapeutics) for the treatment of haemophilia B. In December 2023, fidanacogene elaparvovec received its first approval for the treatment of adults (aged ≥ 18 years) with moderately severe to severe haemophilia B (congenital factor IX deficiency) who are negative for neutralizing antibodies to variant AAV serotype Rh74 (AAVRh74var). Fidanacogene elaparvovec is under regulatory review in the USA and the European Union and clinical studies are ongoing in multiple countries. This article summarizes the milestones in the development of fidanacogene elaparvovec leading to this first approval for moderately severe to severe (factor IX activity $\leq 2\%$) haemophilia B who are negative for neutralizing antibodies to AAVRh74var.

Gene Therapy with Fidanacogene Elaparvovec in Adults with Hemophilia B.

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Fidanacogene elaparvovec, an adeno-associated virus (AAV) gene-therapy vector for hemophilia B containing a high-activity human factor IX variant (FIX-R338L/FIX-Padua), was associated with sustained factor IX activity in a phase 1-2a study. We conducted a phase 3 open-label study of fidanacogene elaparvovec at a dose of 5×10^{11} vector genome copies per kilogram of body weight. Men 18 to 65 years of age with hemophilia B and a factor IX level of 2% or less were eligible for screening if they had received at least 6 months of therapy with prophylactic factor IX concentrate. The primary end point, tested for noninferiority, was the annualized bleeding rate (treated and untreated bleeding episodes) from week 12 to month 15 after treatment with fidanacogene elaparvovec as compared with the prophylaxis lead-in period. Superiority, additional efficacy end points, and safety were also assessed. Of 316 men who underwent screening for the lead-in study, 204 (64.6%) were not eligible; 188 (59.5%) of those were ineligible owing to the presence of anti-AAV neutralizing antibodies. Of the 45 participants who received fidanacogene elaparvovec, 44 completed at least 15 months of follow-up. The annualized rate of bleeding for all bleeding episodes decreased by 71%, from 4.42 (95% confidence interval [CI], 1.80 to 7.05) at baseline to 1.28 (95% CI, 0.57 to 1.98) after gene therapy, a treatment difference of -3.15 episodes (95% CI, -5.46 to -0.83; $P = 0.008$). This result shows the noninferiority and superiority of fidanacogene elaparvovec to prophylaxis. At 15 months, the mean factor IX activity was 26.9% (median, 22.9%; range, 1.9 to 119.0) by one-stage SynthASil assay. A total of 28 participants (62%) received glucocorticoids for increased aminotransferase levels or decreased factor IX levels (or both) starting between 11 and 123 days. No infusion-related serious adverse events, thrombotic events, development of factor IX inhibitors, or malignant conditions were observed. Fidanacogene elaparvovec was superior to

prophylaxis for the treatment of participants with hemophilia B, leading to reduced bleeding and stable factor IX expression. (Funded by Pfizer; BENEGENE-2 ClinicalTrials.gov number, NCT03861273.).
